

Output13 exe13-Bloggerapp.ReactJS-HOL

```
1 import React from "react";
2
3 function CourseDetails() {
4
5   const courses = [
6     { id: 1, course: "Java Full Stack", duration: "3 Months" },
7     { id: 2, course: "React JS", duration: "2 Months" },
8     { id: 3, course: "Spring Boot", duration: "2 Months" }
9   ];
10
11   return (
12     <div>
13       <h2>Course Details</h2>
14       <ul>
15         {courses.map((course) => (
```

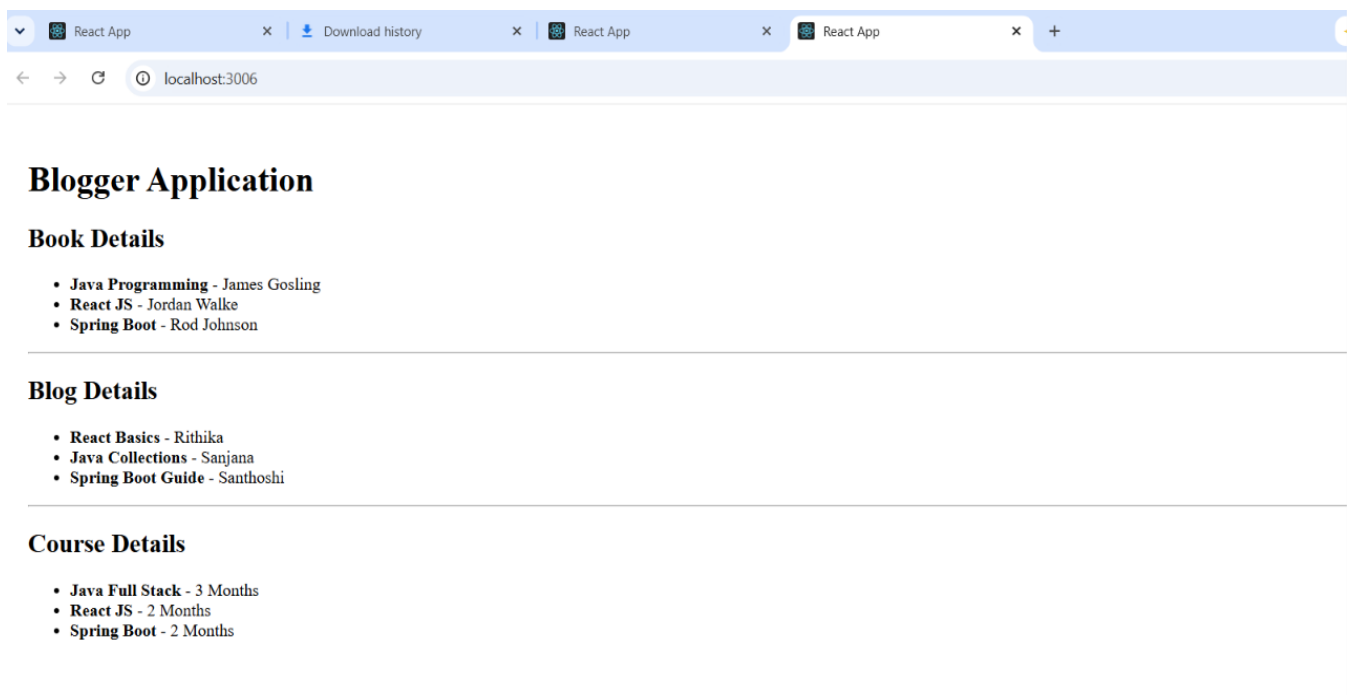
(node:32368) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
(node:32368) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBeforeSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
Starting the development server...
Compiled successfully!

You can now view exe13-bloggerapp in the browser.

Local: http://localhost:3006
On Your Network: http://192.168.0.5:3006

Note that the development build is not optimized.
To create a production build, use `npm run build`.

webpack compiled successfully



Objectives

1. Explain Various Ways of Conditional Rendering

Conditional rendering in React allows components or UI elements to be displayed based on specific conditions. It helps create dynamic applications that respond to user actions and application state.

Ways of Conditional Rendering

- Using if...else statements
- Using the Ternary Operator (condition ? true : false)
- Using the Logical AND (&&) Operator
- Returning null to prevent rendering

2. Explain How to Render Multiple Components

React allows multiple components to be rendered together by including them inside a parent component. This promotes modularity, code reuse, and better organization of the application.

Benefits

- Improves code reusability.
- Makes applications modular.
- Simplifies maintenance.

3. Define List Component

A List Component is used to display multiple items dynamically from an array of data. React commonly uses the `map()` function to generate list elements efficiently.

4. Explain About Keys in React Applications

A key is a unique identifier assigned to each element in a list. Keys help React identify which items have changed, been added, or removed, allowing efficient updates to the user interface.

Why Are Keys Important?

- Improve rendering performance.
- Help React efficiently update the DOM.
- Prevent unnecessary re-rendering.
- Ensure each list item is uniquely identified.

Best Practice: Use a unique ID from your data instead of the array index whenever possible.

5. Explain How to Extract Components with Keys

When rendering lists, each item can be moved into its own reusable component. The key should be assigned when the component is created in the list, not inside the child component.

Benefits

- Makes components reusable.
- Keeps code clean and organized.
- Improves maintainability.

6. Explain React map() Function

The `map()` function is a JavaScript array method widely used in React to transform each element of an array into a React element. It returns a new array without modifying the original array.